

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)

Question Count: 5

Total Marks: 50

Code of Conduct

I A. Reddy Prasad (192472045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _A.Reddy Prasad

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter

Answer

A **hypervisor** is software that creates and manages Virtual Machines (VMs).

Type-1 Hypervisor (Bare-Metal)

- Runs directly on the physical hardware.
- Does not require a host operating system.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen.
- Provides high performance and strong security.

Type-2 Hypervisor (Hosted)

- Runs on top of a host operating system.
- Easier to install and use.
- Examples: VirtualBox and VMware Workstation.
- Suitable for testing and development.

Comparison

Feature	Type-1	Type-2
Performance	High	Moderate
Security	High	Lower
Speed	Faster	Slower

Resource Usage Efficient More Overhead

Justification

For a cloud datacenter hosting **mission-critical applications**, **Type-1 Hypervisor** is the better choice because it provides:

- Better performance
- Higher security
- Greater reliability
- Efficient resource utilization
- Enterprise-level management features

Hence, Type-1 hypervisors are widely used in cloud data centers.

2. Virtualization Architecture for Workload Isolation

Answer

Virtualization allows multiple virtual machines to run on a single physical server while keeping them isolated.

Workload Isolation

The finance, healthcare, and student portal applications run in separate VMs. Each VM has:

- Independent operating system
- Separate memory
- Separate storage
- Virtual CPU
- Separate virtual network

The hypervisor prevents one VM from accessing another VM's resources.

Major Attack Surfaces

1. Hypervisor Attack

- If the hypervisor is compromised, attackers may gain access to multiple VMs.

2. VM Escape

- A malicious VM escapes its environment and attacks other VMs or the hypervisor.

Conclusion

Virtualization provides secure workload isolation, efficient resource sharing, and improved security. Proper patching, access control, and monitoring help reduce attack risks.

3. VM Host Utilization Analysis

Answer

Given Data

- CPU Utilization = **92%**
- Memory Utilization = **88%**
- Storage I/O Latency = **35 ms**
- VM Boot Time = **170 seconds**

Analysis

CPU (92%)

- CPU is overloaded.
- Applications may become slow.

Memory (88%)

- Memory usage is very high.
- System may start swapping memory.

Storage Latency (35 ms)

- High latency indicates slow disk performance.
- VMs take longer to access data.

Boot Time (170 s)

- Long boot time is caused by CPU, memory, and storage bottlenecks.

Corrective Actions

- Increase CPU resources.
- Upgrade RAM.
- Use SSD or NVMe storage.
- Balance workloads across multiple hosts.
- Remove unnecessary VMs.

Conclusion

The major bottlenecks are CPU overload, high memory usage, and slow storage. Resource upgrades and load balancing will improve VM performance.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter

Answer

A **Software Defined Datacenter (SDDC)** virtualizes compute, storage, and networking, allowing infrastructure to be managed through software.

Compute Virtualization

- Multiple virtual machines run on one physical server.
- Resources can be allocated dynamically.

Storage Virtualization

- Combines multiple storage devices into one logical storage pool.
- Improves storage flexibility and utilization.

Network Virtualization

- Creates virtual networks independent of physical hardware.
- Enables faster network configuration.

Advantages over Traditional Datacenter

- Faster deployment
- Better scalability
- Automated resource management
- Improved flexibility
- Reduced operational cost

Conclusion

Compared to traditional datacenters, SDDC offers greater agility through automation, virtualization, and centralized management of compute, storage, and networking resources.

5. Multi-Cloud Federation Identity Management

Answer

In a multi-cloud environment, different cloud providers may use different identity systems, causing authentication and access issues.

Problem Analysis

- Different user accounts across providers
- Inconsistent access permissions
- Authentication failures
- Security risks

Proposed Solution

Use **Identity Federation** with:

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Protocols such as **SAML**, **OAuth 2.0**, or **OpenID Connect (OIDC)** allow users to authenticate once and securely access multiple cloud providers.

Privacy and Security

- Encrypt data using AES-256.
- Protect communication using TLS/SSL.
- Maintain audit logs for monitoring.
- Apply least-privilege access through RBAC.

Conclusion

Identity federation improves security, simplifies user management, and enables secure access across multiple cloud providers while maintaining privacy and compliance.